

Jetpack old system upgraded to SUPER system

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1、ReadMe

This tutorial applies only to upgradeable Jetson Orin configurations; users with Jetson Nano or Raspberry Pi versions should skip this tutorial.

This tutorial is primarily intended for the following customers:

1. Users who purchased a Jetson Orin Nano/NX prior to March 31, 2025.

2、Write SUPER boot file

The purpose of this tutorial is to write SUPER boot to the Jetson Orin series board (for use with Jetpack 6.2 system).

The process of writing the boot file does not require the install SSD.

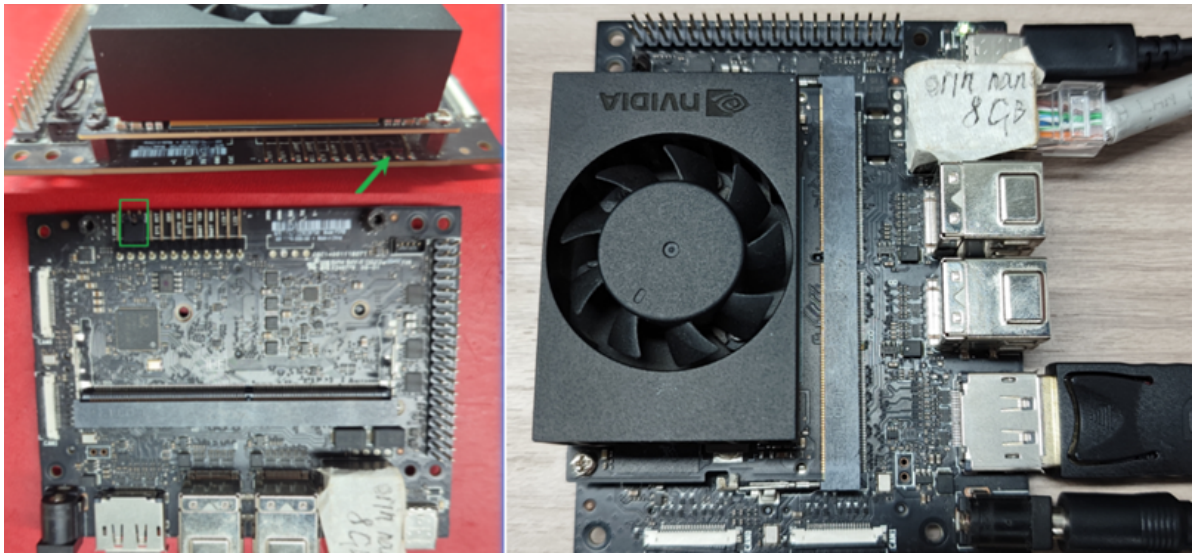
2.1、Connect hardware

1. Use a jumper cap to short-circuit the **FC REC** and **GND** pins under the core board. You don't need to remove the core board, the picture is just for a clearer observation

This diagram takes Jetson Orin Nano official version as an example, and users of other versions can refer to it for use (the hardware interface and functional layout are consistent)

2. The Jetson Orin board needs to be connected to a DC power adapter, DP data cable, and network cable. And connect it to the computer via a Type C data cable

Note: You don't need to connect the DP data cable and network cable when writing the boot, but they will be needed when using the motherboard later.



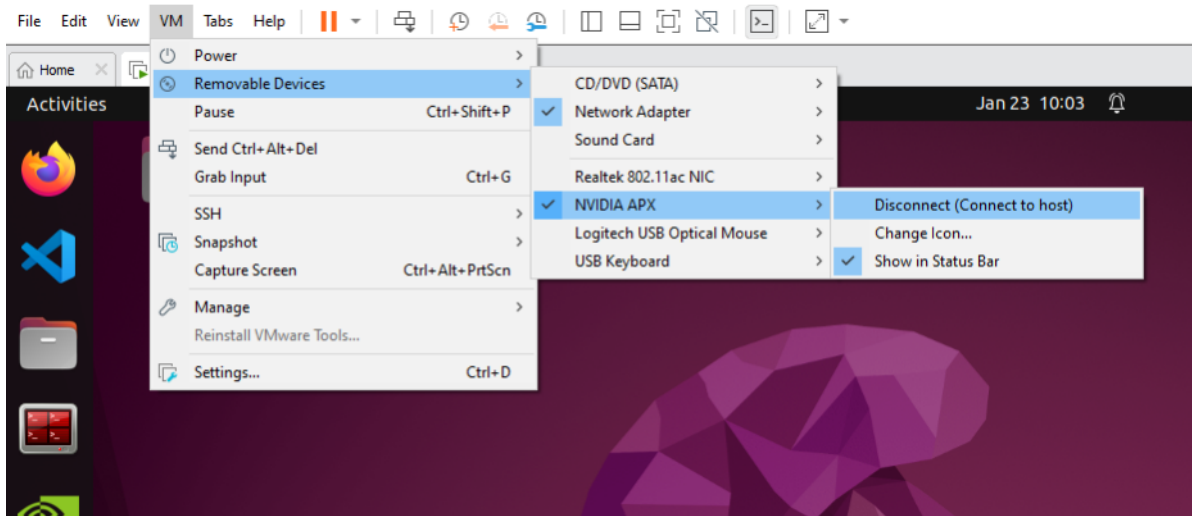
2.2、 Connect VM software

Users need to use the virtual machine file provided by Yahboom to write the SUPER boot file.

We need to connect the motherboard to the virtual machine so that it can be recognized by the Ubuntu system.

User name: yahboom

Password: yahboom



```
yahboom@VM: ~
yahboom@VM: ~ 118x30
yahboom@VM:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 006: ID 0955:7523 NVIDIA Corp. APX
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
yahboom@VM:~$
```

2.3、 Write SUPER boot file

Open the terminal, enter the specified folder and run the script.

If writing fails, you can disconnect the motherboard power supply, reconnect the virtual machine to run the command, and try again

```
cd ~/jetpack_6.2/Linux_for_Tegra && sudo ./yahboom_flash.sh.x
```

```
yahboom@VM: ~
yahboom@VM: ~ 118x30
yahboom@VM:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 006: ID 0955:7523 NVIDIA Corp. APX
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
yahboom@VM:~$ cd ~/jetpack_6.2/Linux_for_Tegra && sudo ./yahboom_flash.sh.x
[sudo] password for yahboom:
#####
# L4T BSP Information:
# R36 , REVISION: 4.3
# User release: 0.0
#####
ECID is 0x80012344705DF1D76C0000001FD8140
# Target Board Information:
# Name: jetson-orin-nano-devkit-super, Board Family: generic, SoC: Tegra 234,
# OpMode: production, Boot Authentication: NS,
# Disk encryption: disabled ,
#####
copying device_config(/home/yahboom/jetpack_6.2/Linux_for_Tegra/bootloader/generic/BCT/tegra234-mb1-bct-device-p3767-0
000.dts)... done.
copying misc_config(/home/yahboom/jetpack_6.2/Linux_for_Tegra/bootloader/generic/BCT/tegra234-mb1-bct-misc-p3767-0000.
dts)... done.
copying emc_fuse_dev_params(/home/yahboom/jetpack_6.2/Linux_for_Tegra/bootloader/generic/BCT/tegra234-br-bct-diag-boot
.dts)... done.
copying minratchet_config(/home/yahboom/jetpack_6.2/Linux_for_Tegra/bootloader/generic/BCT/tegra234-mb1-bct-ratchet-p3
```

```
yahboom@VM: ~/jetpack_6.2/Linux_for_Tegra
yahboom@VM: ~/jetpack_6.2/Linux_for_Tegra 135x35
[ 76.0156 ] Writing partition BCT with br_bct_BR.bct [ 8192 bytes ]
[ 76.0160 ] [.....] 100%
[ 82.6710 ] tegradevflash_v2 --write BCT-boot-chain_backup bct_backup.img
[ 82.6717 ] Bootloader version 01.00.0000
[ 83.0245 ] Writing partition BCT-boot-chain_backup with bct_backup.img [ 32768 bytes ]
[ 83.0248 ] [.....] 100%
[ 83.4461 ] tegradevflash_v2 --write A_MB1_BCT mb1_cold_boot_bct_MB1_sigheader.bct.encrypt
[ 83.4467 ] Bootloader version 01.00.0000
[ 83.7399 ] Writing partition A_MB1_BCT with mb1_cold_boot_bct_MB1_sigheader.bct.encrypt [ 17600 bytes ]
[ 83.7402 ] [.....] 100%
[ 83.9829 ] tegradevflash_v2 --write B_MB1_BCT mb1_cold_boot_bct_MB1_sigheader.bct.encrypt
[ 83.9833 ] Bootloader version 01.00.0000
[ 84.2747 ] Writing partition B_MB1_BCT with mb1_cold_boot_bct_MB1_sigheader.bct.encrypt [ 17600 bytes ]
[ 84.2754 ] [.....] 100%
[ 84.5188 ] tegradevflash_v2 --write A_MEM_BCT mem_coldboot_sigheader.bct.encrypt
[ 84.5192 ] Bootloader version 01.00.0000
[ 84.8233 ] Writing partition A_MEM_BCT with mem_coldboot_sigheader.bct.encrypt [ 243712 bytes ]
[ 84.8236 ] [.....] 100%
[ 87.8696 ] tegradevflash_v2 --write B_MEM_BCT mem_coldboot_sigheader.bct.encrypt
[ 87.8703 ] Bootloader version 01.00.0000
[ 88.1617 ] Writing partition B_MEM_BCT with mem_coldboot_sigheader.bct.encrypt [ 243712 bytes ]
[ 88.1632 ] [.....] 100%
[ 91.2087 ] Flashing completed

[ 91.2088 ] Coldbooting the device
[ 91.2096 ] tegrarcv_v2 --chip 0x23 0 --isnb2
[ 91.2101 ] MB2 version 01.00.0000
[ 91.5296 ] Coldbooting the device
[ 91.5301 ] tegrarcv_v2 --chip 0x23 0 --reboot coldboot
[ 91.5306 ] MB2 version 01.00.0000
*** The target generic has been flashed successfully. ***
Reset the board to boot from internal eMMC.

Total:132 seconds
yahboom@VM:~/jetpack_6.2/Linux_for_Tegra$
```

3、Write Yahboom robot system file into SSD

This tutorial allows you to write the system image file of the robot car SUPER (Jetpack6.2) provided by Yahboom to the SSD

Note: Users need to prepare the M.2 SSD Enclosure in advance, install the SSD into the M.2 SSD Enclosure, connect it to the computer, format it and write the new system image file.

3.1、Format SSD

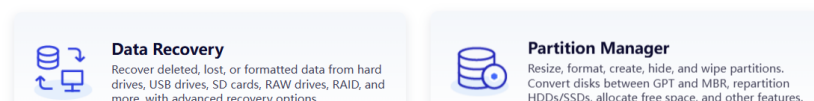
Before writing the Yahboom robot car system image file, you need to format the solid-state drive into exFAT format.

3.1.1、Download DiskGenius

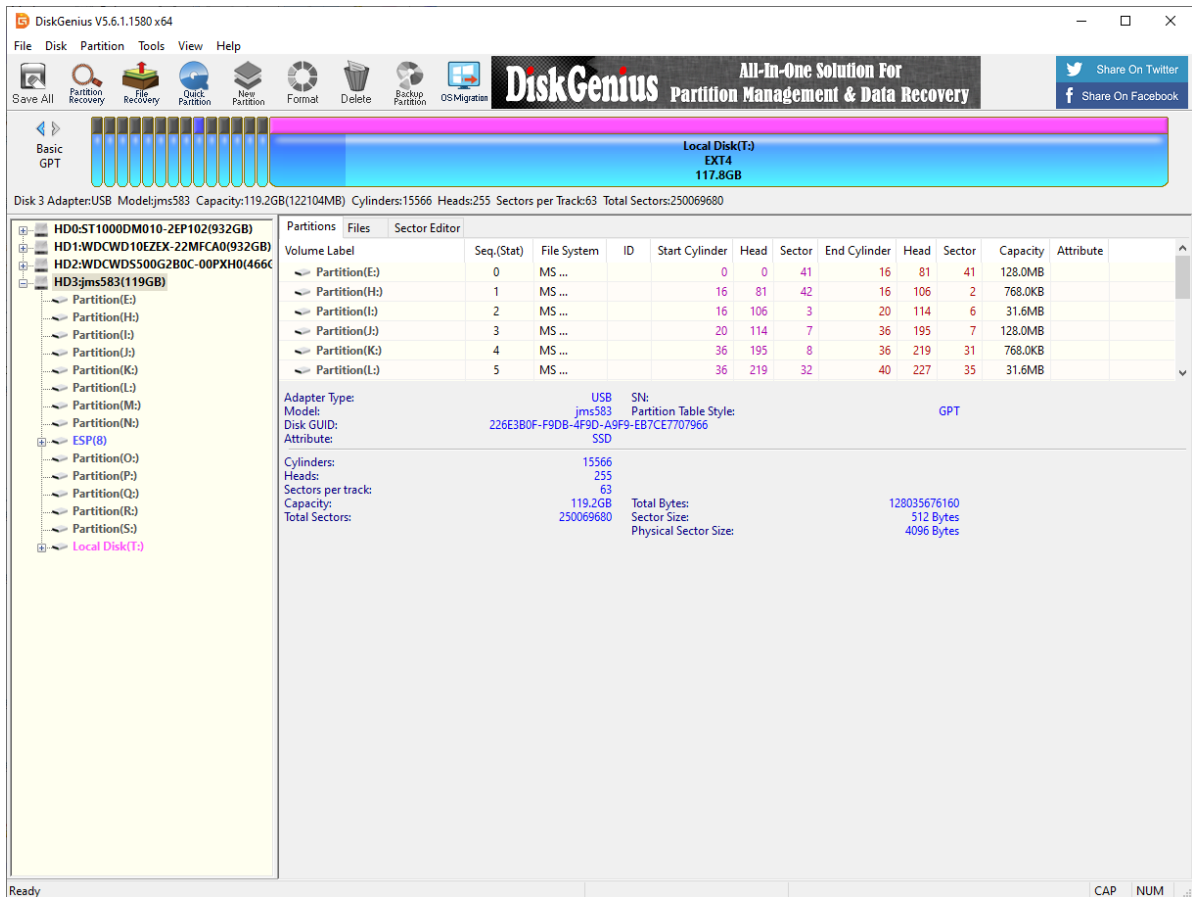
Download link: <https://www.diskgenius.com/>



What Can DiskGenius Do?



Unzip the downloaded compressed file and open `DiskGenius.exe` to use.



3.1.2、Use DiskGenius

Delete partition

Deleting a partition will clear the disk data. Please confirm that the drive letter is the disk that needs to be formatted before confirming the operation.

You can judge based on the disk size and the newly added drive letter of the connected disk.

DiskGenius V5.6.1.1580 x64

File Disk Partition Tools View Help

Save All Partition Recovery File Recovery Quick Partition New Partition Format Delete Backup Partition OS Migration

DiskGenius All-In-One Solution For Partition Management & Data Recovery

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Basic GPT

Local Disk(T:) EXT4 117.8GB

Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680

Partitions Files Sector Editor

Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute

0	MS ...		0	0	41	16	81	41	128.0MB	
1	MS ...		16	81	42	16	106	2	768.0KB	
2	MS ...		16	106	3	20	114	6	31.6MB	
3	MS ...		20	114	7	36	195	7	128.0MB	
4	MS ...		36	195	8	36	219	31	768.0KB	
5	MS ...		36	219	32	40	227	35	31.6MB	

USB jms583 SN: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966 Partition Table Style: GPT SSD

15566 255 63 119.2GB 250069680 Total Bytes: 128035676160 Sector Size: 512 Bytes Physical Sector Size: 4096 Bytes

Save Partition Table(F8) Backup Partition Table(F2) Restore Partition Table(F3) Search Lost Partitions(Partition Recovery)(L) Recover Lost Files(U) Rebuild Master Boot Record(M) Clear Reserved Sectors(E) Quick Partition(F6) **Delete All Partitions(A)** Erase Sectors(E) Backup Disk To Image File Restore Image File To Disk

Convert To GUID Partition Table (P) Convert To MBR Partition Table (B) Convert To Basic Disk (D) Appoint Disk Geometry(G) Modify MBR Signature / Disk GUID View S.M.A.R.T. Information(S) View Storage Pool Information Verify Or Repair Bad Sectors(Y) Reset Bad Sector Records

Convert Boot Mode Change Device State TRIM Optimization

Close Virtual Disk File(C) Recompose Virtual RAID

Delete all partitions

DiskGenius V5.6.1.1580 x64

File Disk Partition Tools View Help

Save All Partition Recovery File Recovery Quick Partition New Partition Format Delete Backup Partition OS Migration

DiskGenius All-In-One Solution For Partition Management & Data Recovery

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Basic GPT

Local Disk(T:) EXT4 117.8GB

Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680

Partitions Files Sector Editor

Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute

Partition(E:)	0	MS ...		0	0	41	16	81	41	128.0MB	
Partition(H:)	1	MS ...		16	81	42	16	106	2	768.0KB	
Partition(I:)	2	MS ...		16	106	3	20	114	6	31.6MB	
Partition(J:)	3	MS ...		20	114	7	36	195	7	128.0MB	
Partition(K:)	4	MS ...		36	195	8	36	219	31	768.0KB	
Partition(L:)	5	MS ...		36	219	32	40	227	35	31.6MB	

Partition(E:)

Partition(H:)

Partition(I:)

Partition(J:)

Partition(K:)

Partition(L:)

Partition(M:)

Partition(N:)

ESP(8)

Partition(O:)

Partition(P:)

Partition(Q:)

Partition(R:)

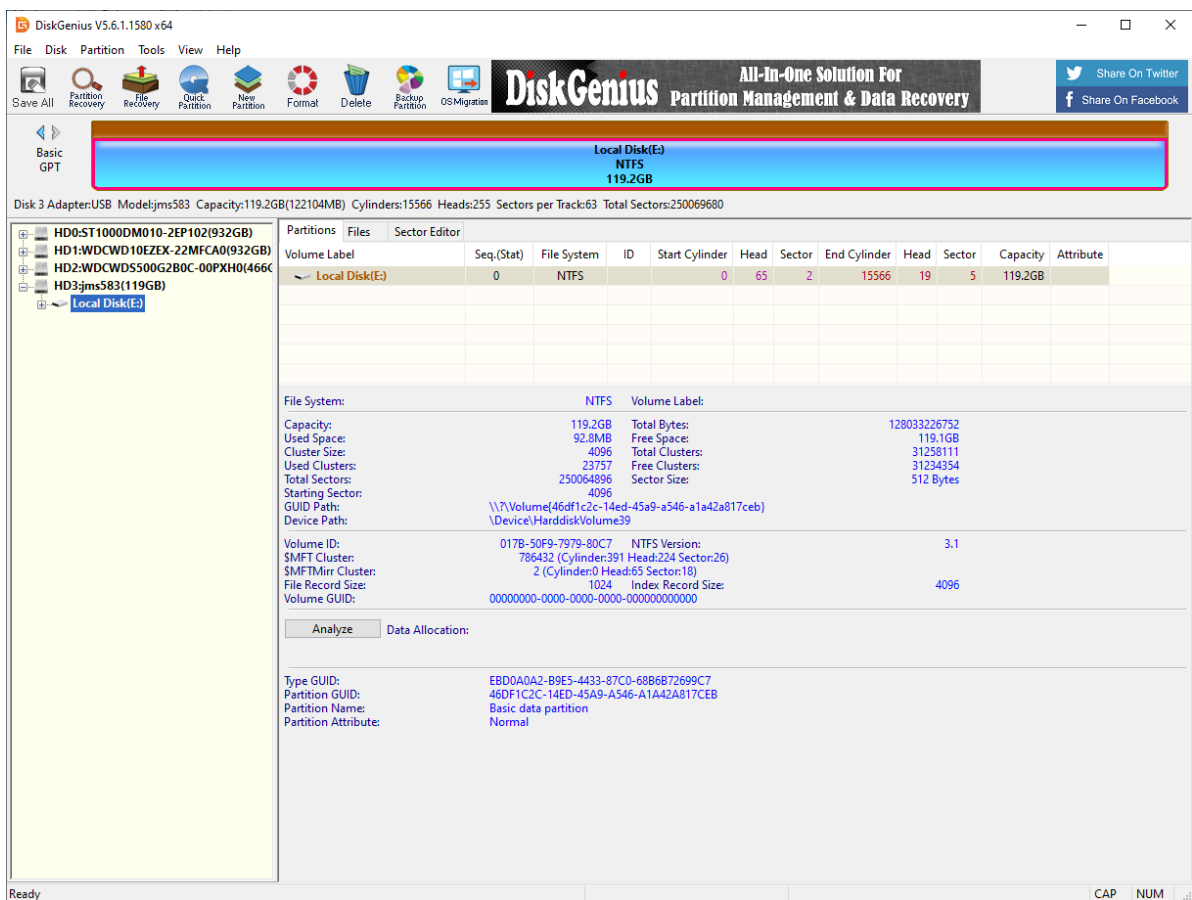
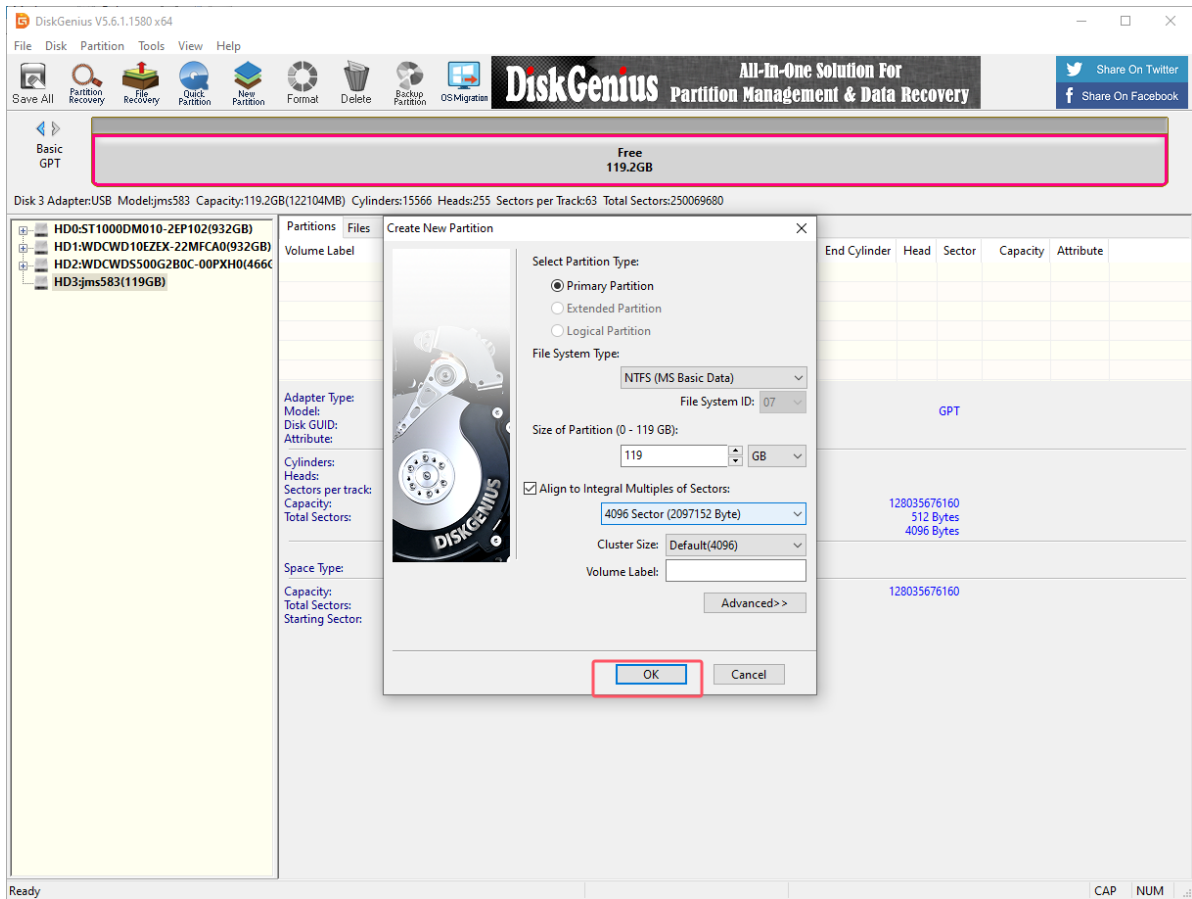
Partition(S:)

Local Disk(T:)

Are you sure you want to delete all the partitions on current disk? All files in it will lose. Select "Yes" to delete.

Yes No

Ready

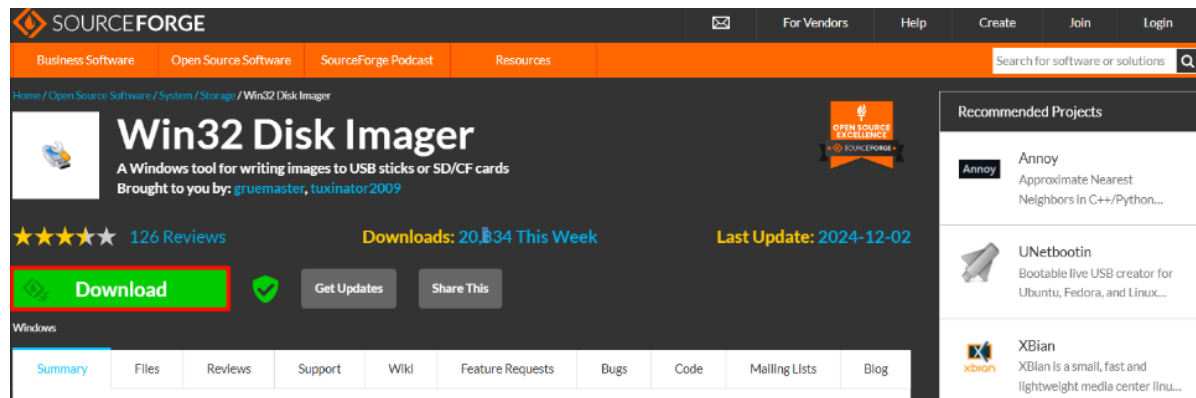


3.2、Write Yahboom robot car system file

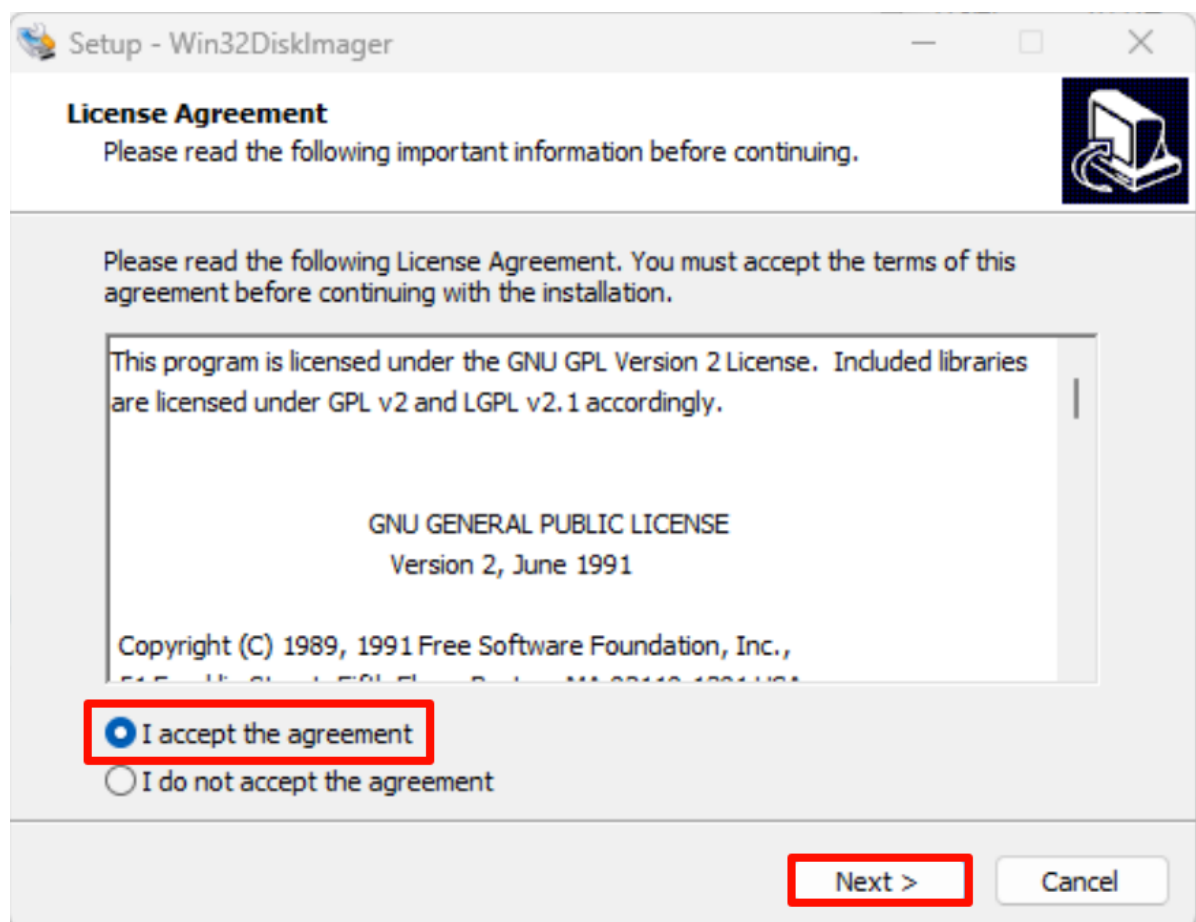
You need to download the image system file in the tutorials in advance and decompress it to get the .img file. Download the system and decompress it locally.

3.2.1, Install Win32DiskImager

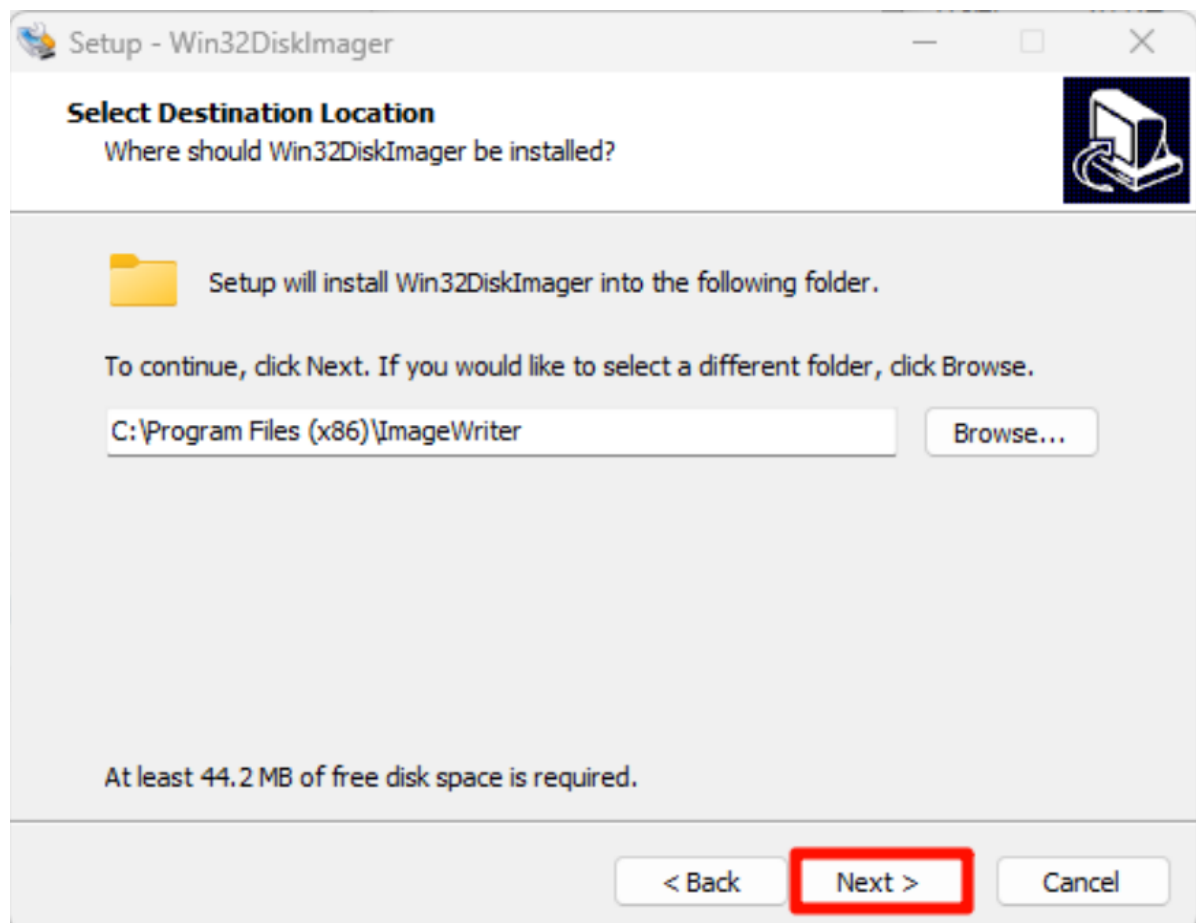
Download link: <https://sourceforge.net/projects/win32diskimager/>



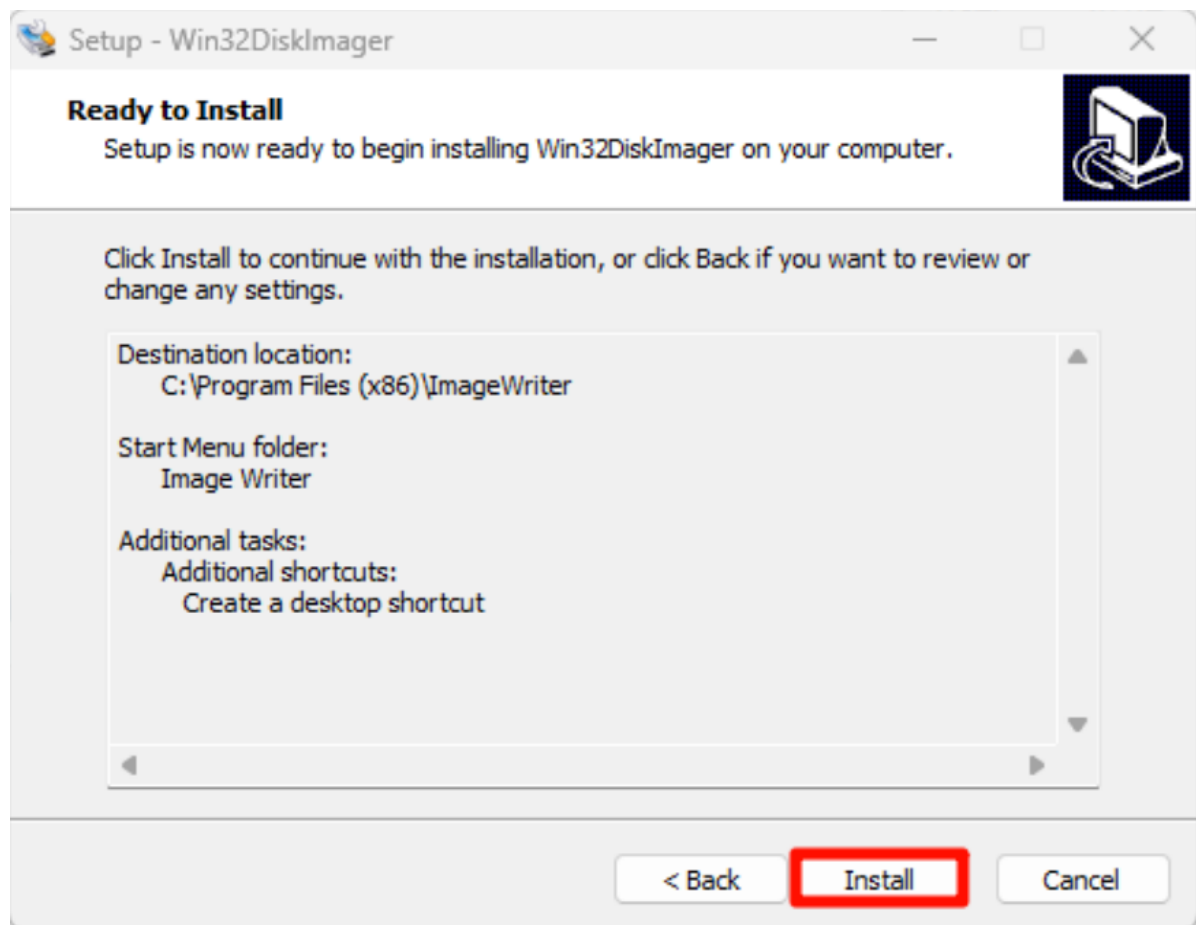
Open the `win32diskimager-1.0.0-install.exe` installation package as an administrator and accept the agreement.

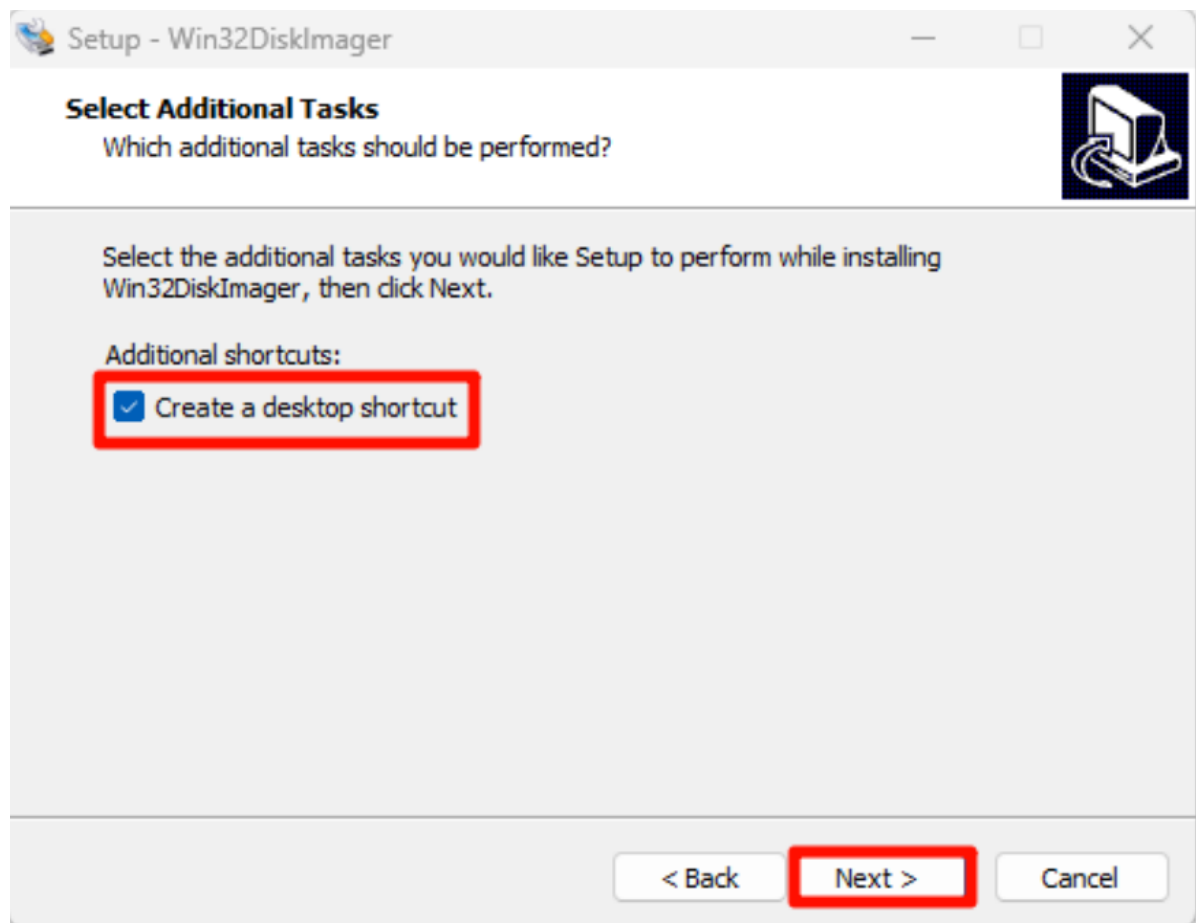


Installation location: Default location is recommended.

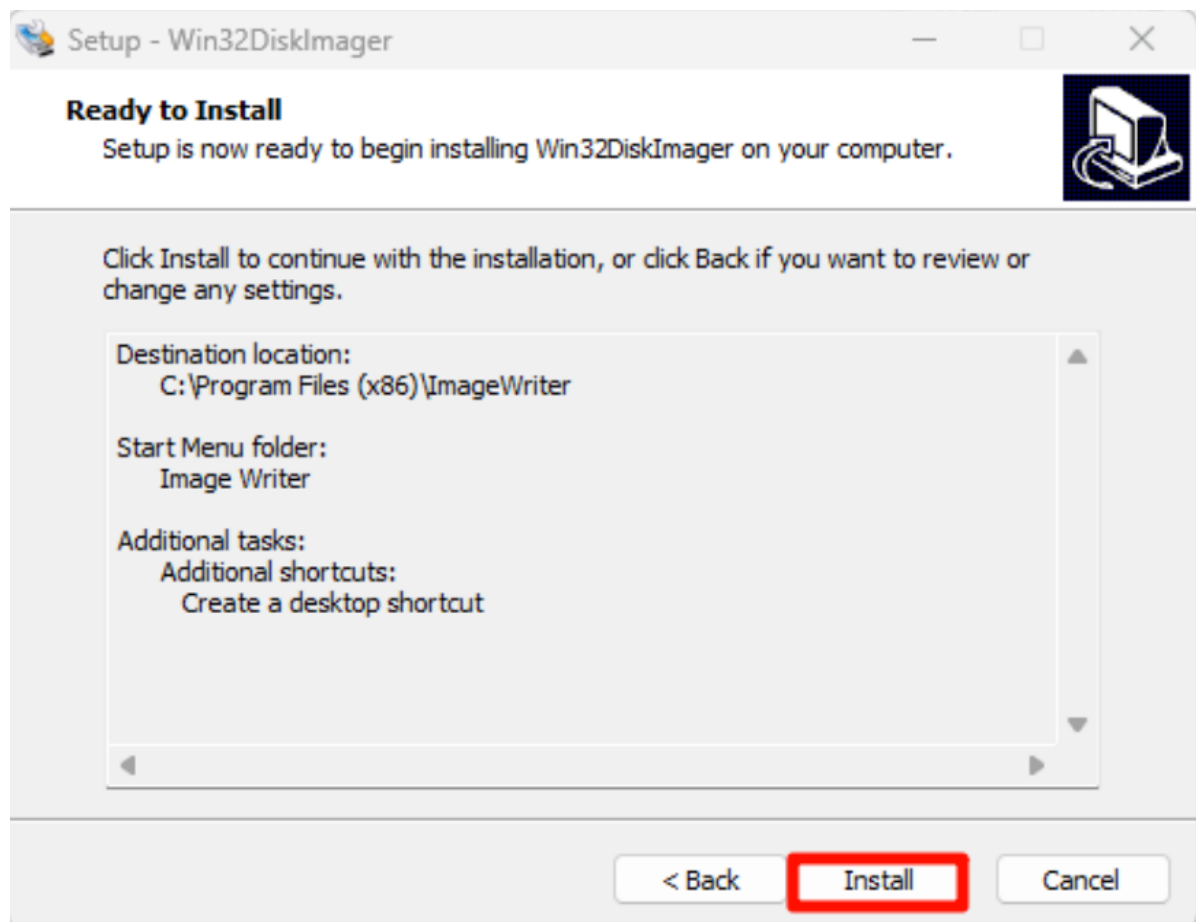


Click " Install"

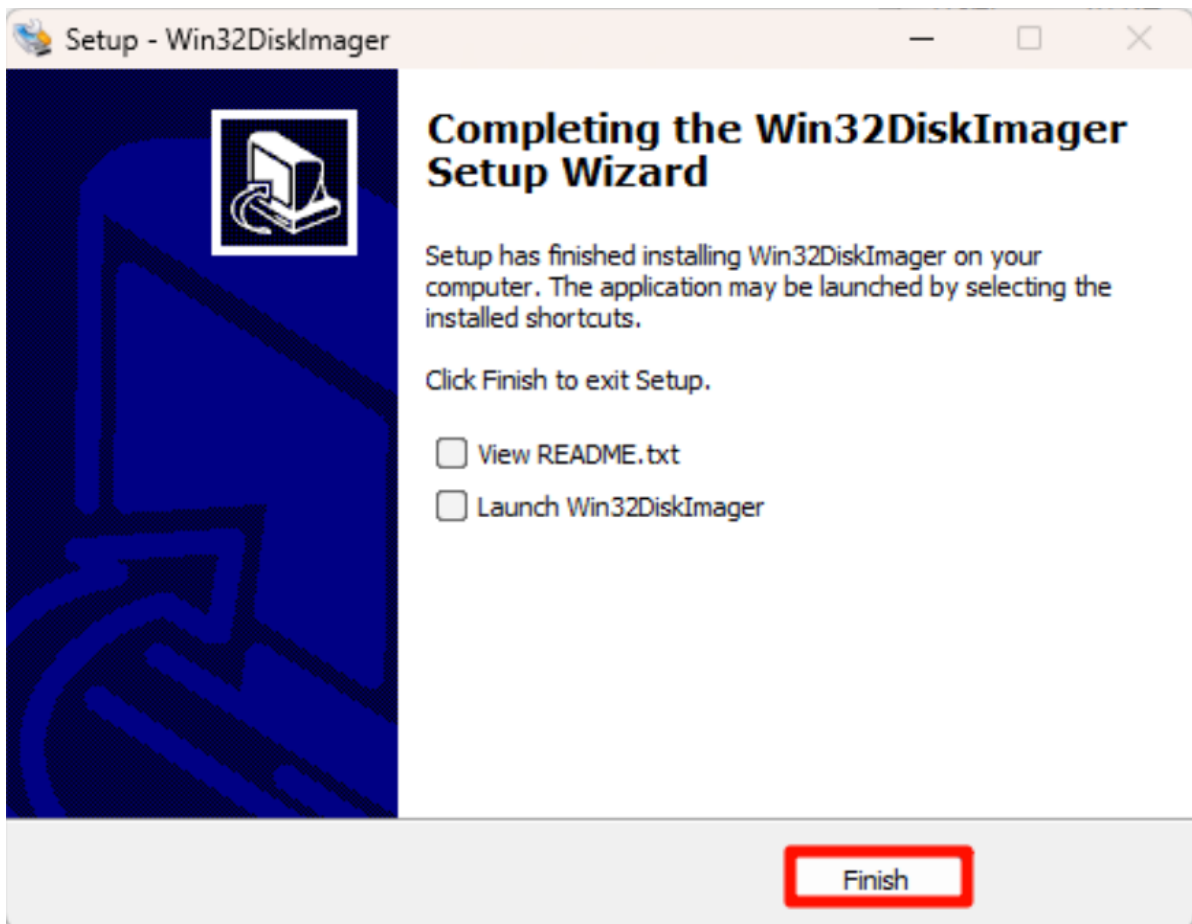




Click " Install"

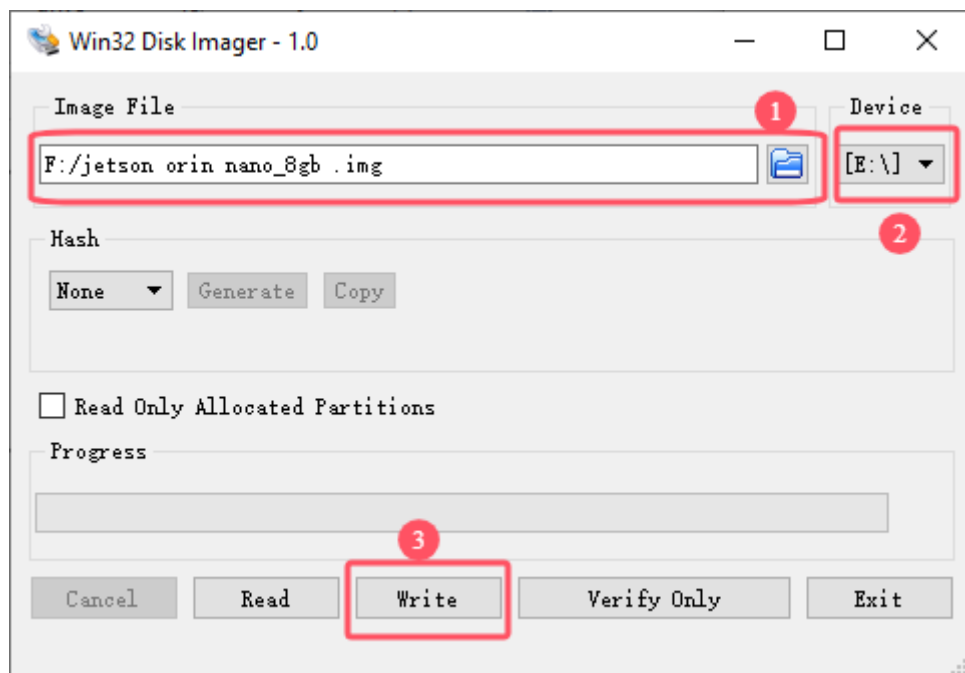


Complete the installation.

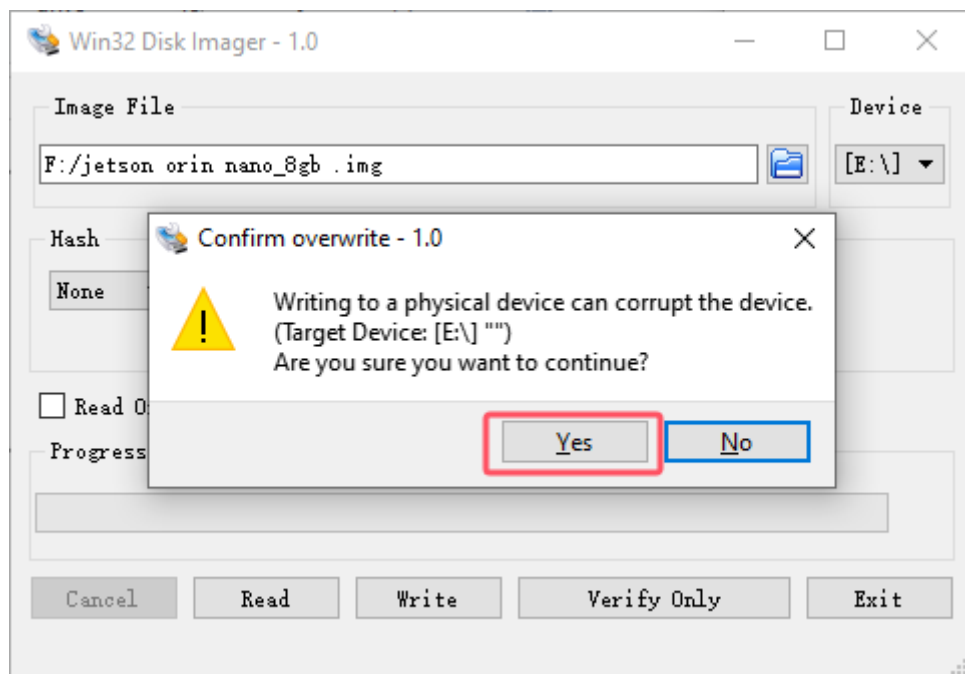


3.2.2. Use Win32DiskImager

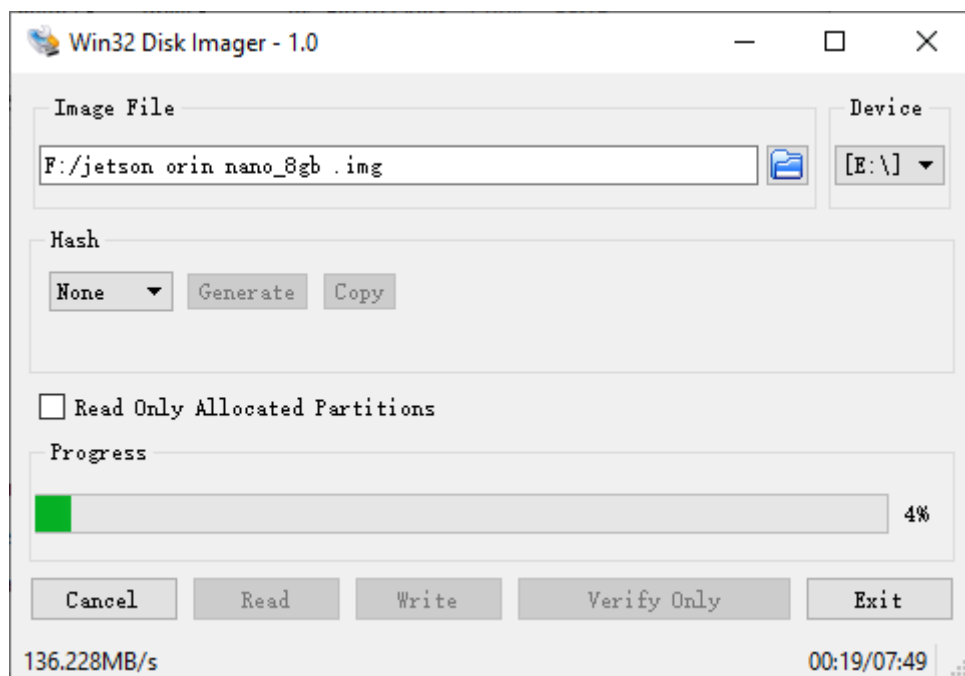
- ①: Select the factory image file (*.img) in the data
- ②: Select the drive letter corresponding to the SSD
- ③: Write the Yahboom system image to the SSD

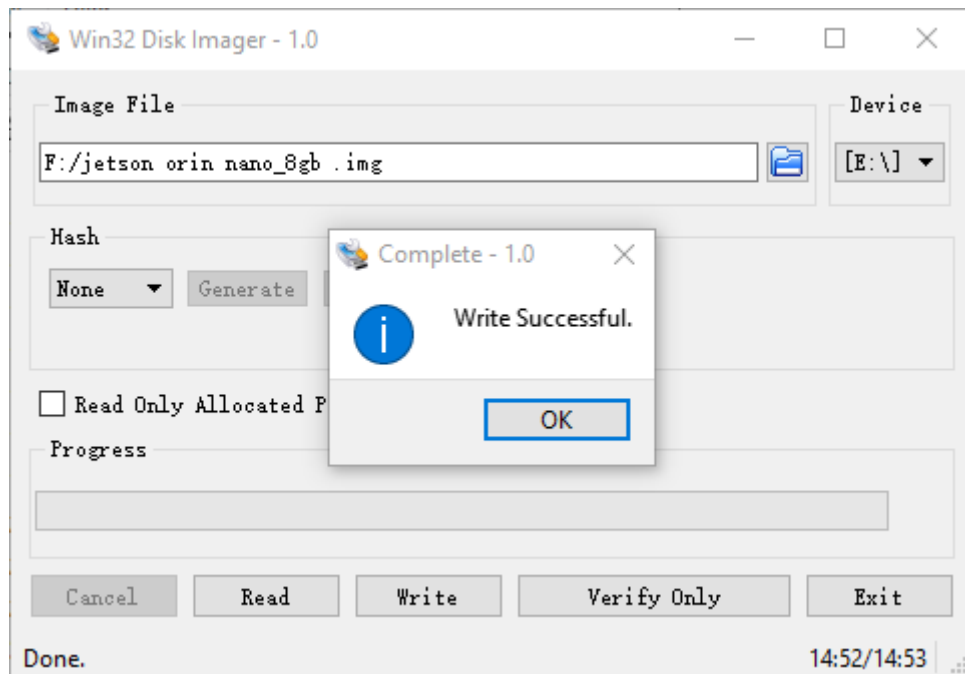


Click "Yes"



wait it





After the system is written, you can close this software.

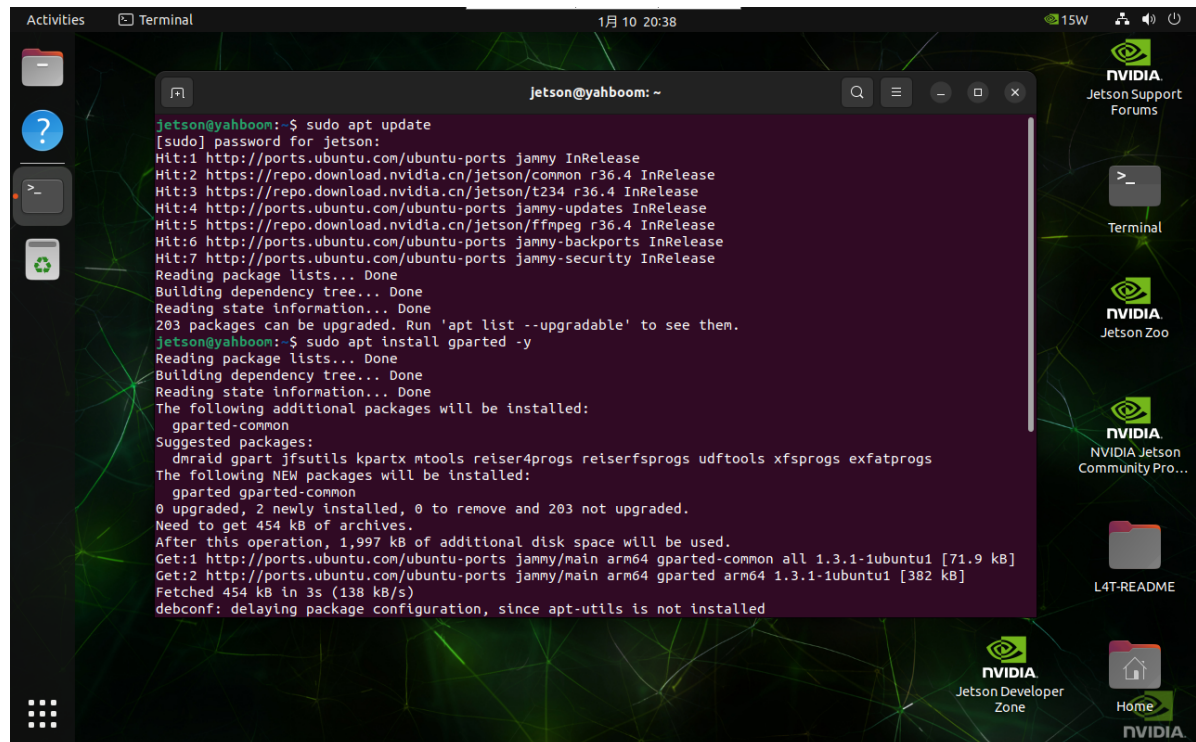
And install the SSD to the Jetson Orin board.

4、SSD expansion

This tutorial can be used to allocate the unallocated space of the solid state drive.

4.1、Install GParted

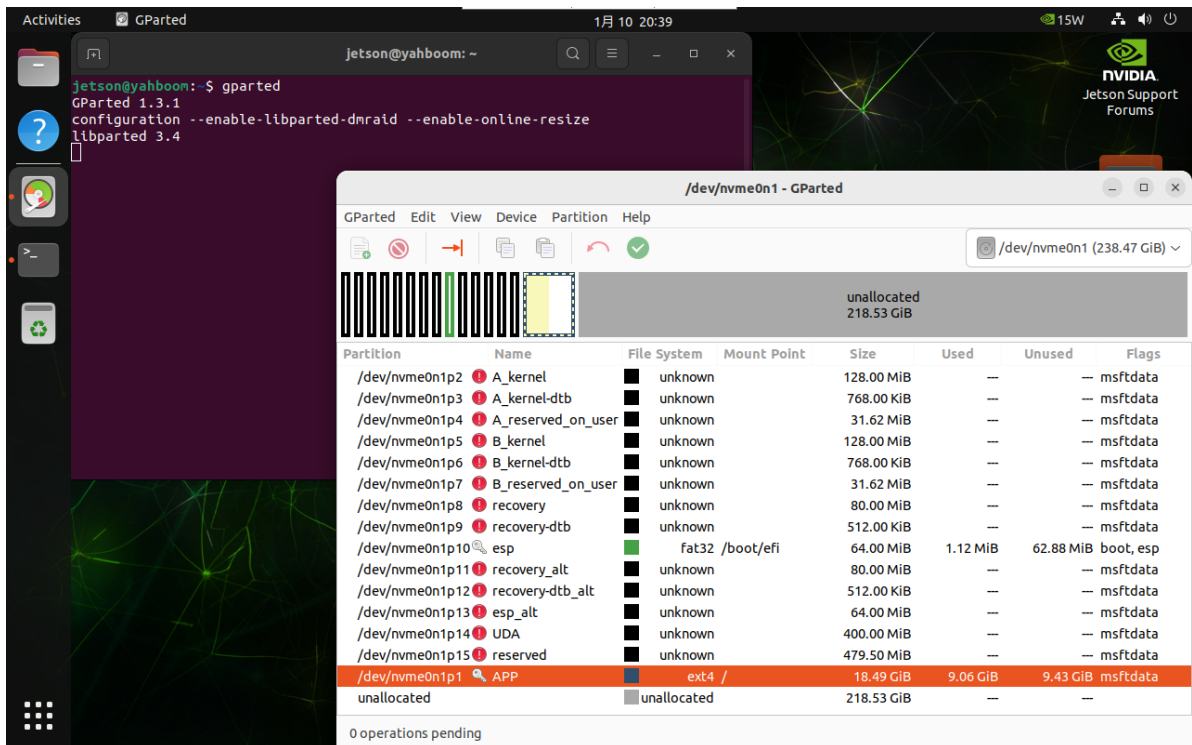
```
sudo apt update
sudo apt install gparted -y
```



4.2. Use GParted

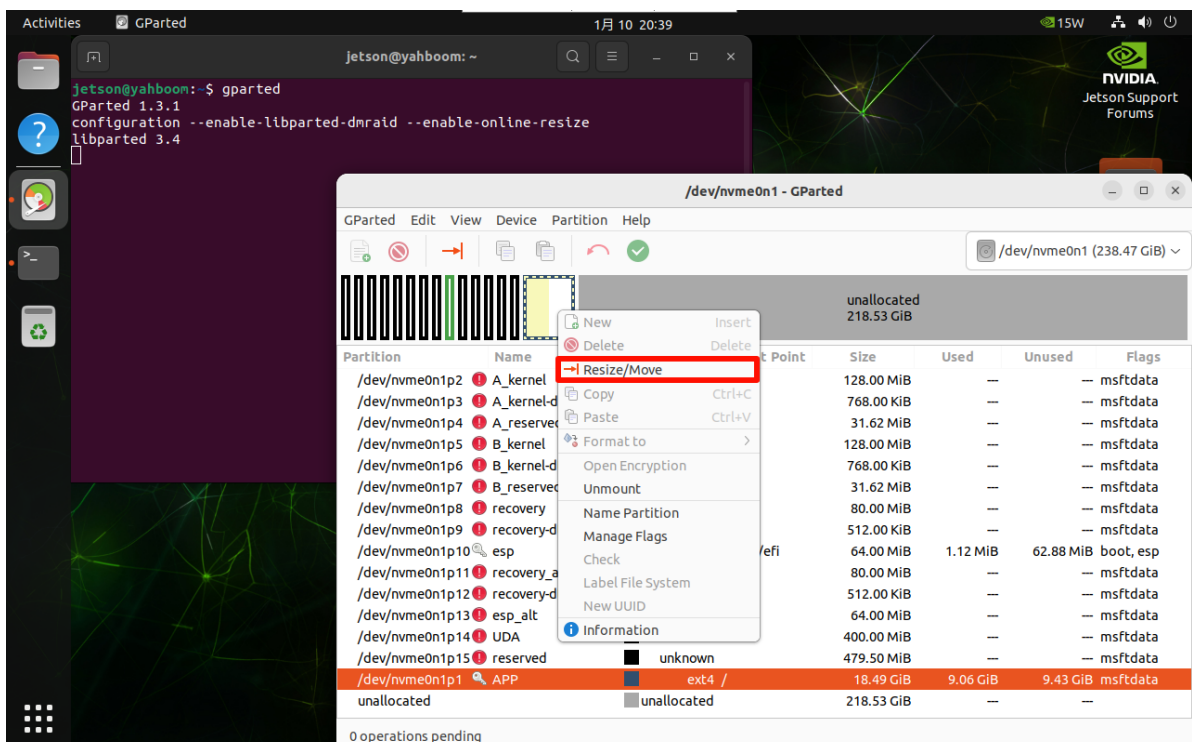
Find the **GParted** application icon in the system application menu bar to open it or enter the following command in the terminal to start it.

```
gparted
```

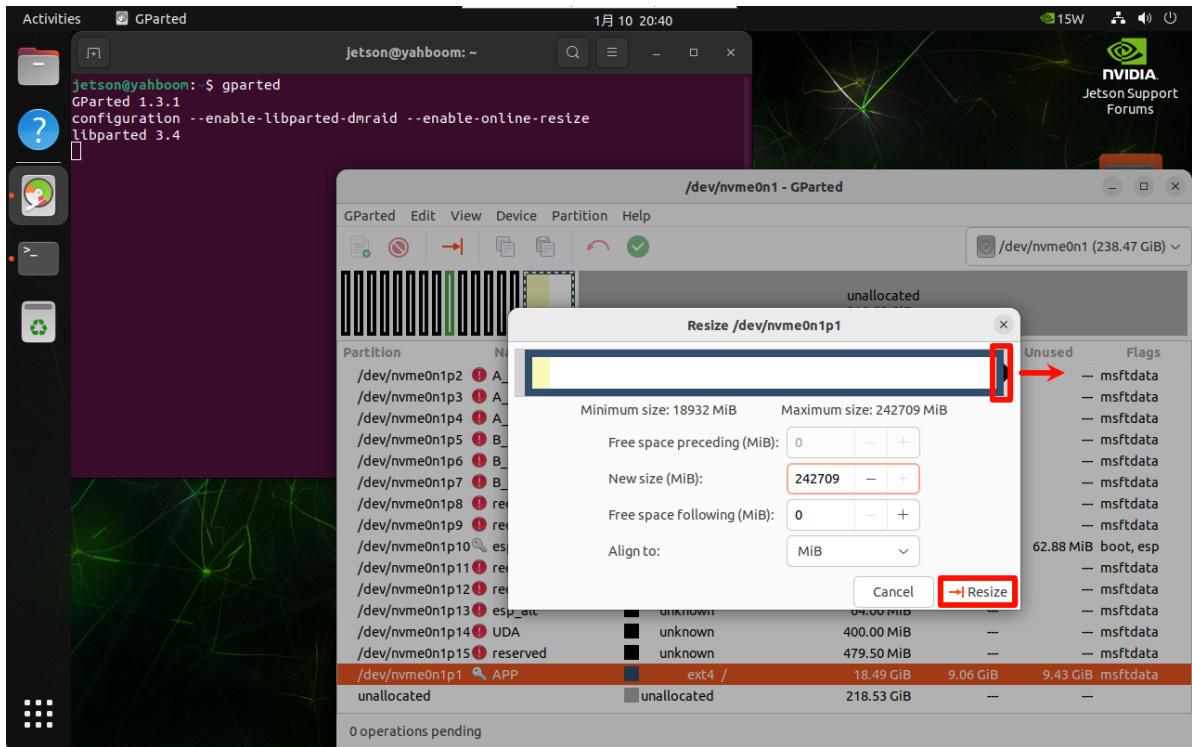


4.3. Adjust partitions

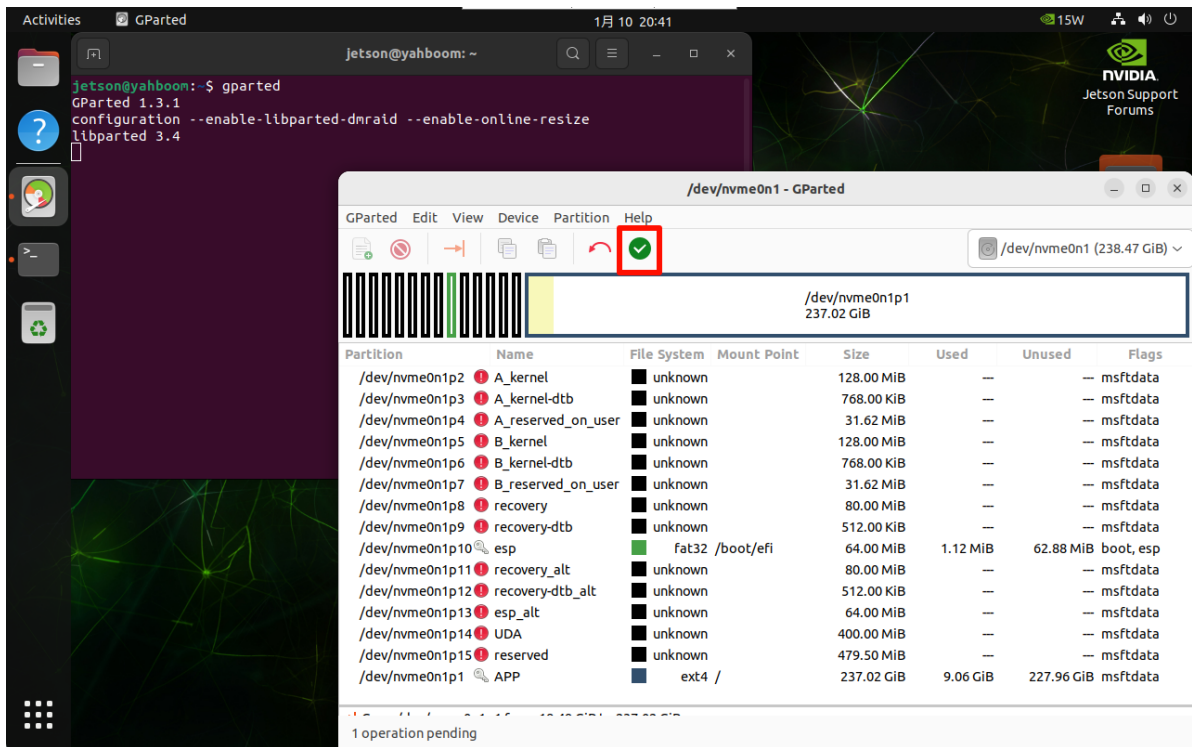
Right-click the disk partition that needs to be expanded: generally select the largest partition on the disk

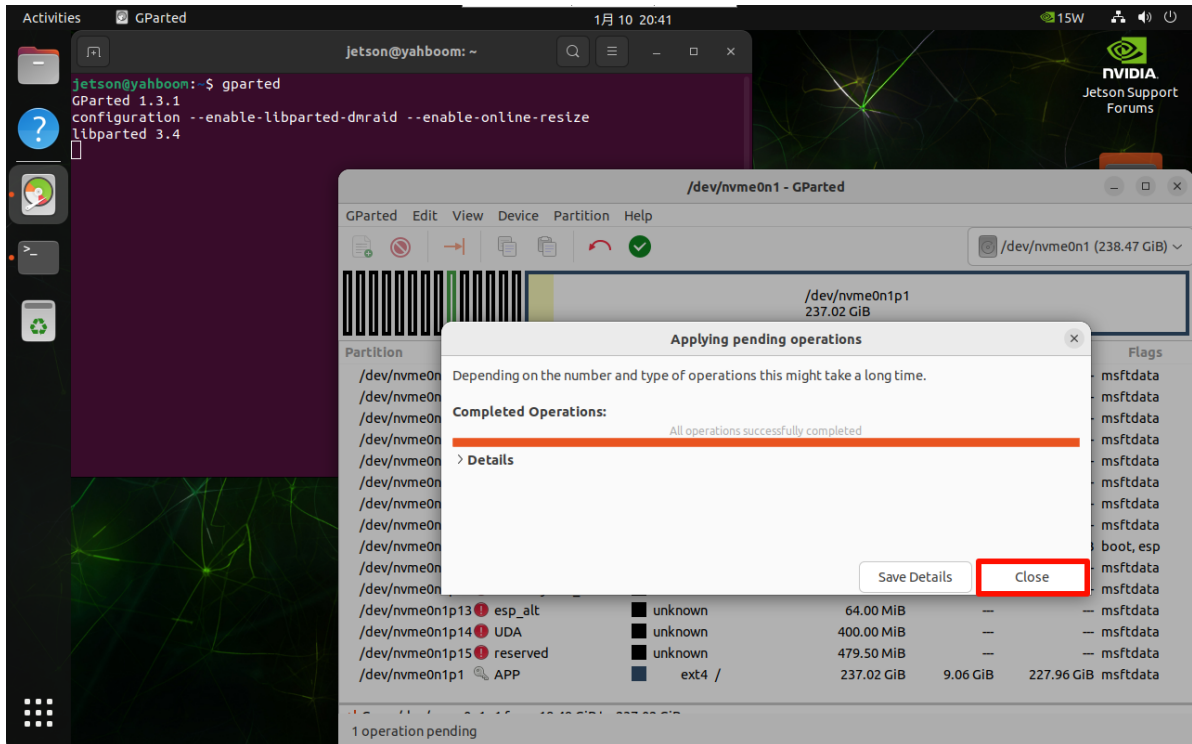
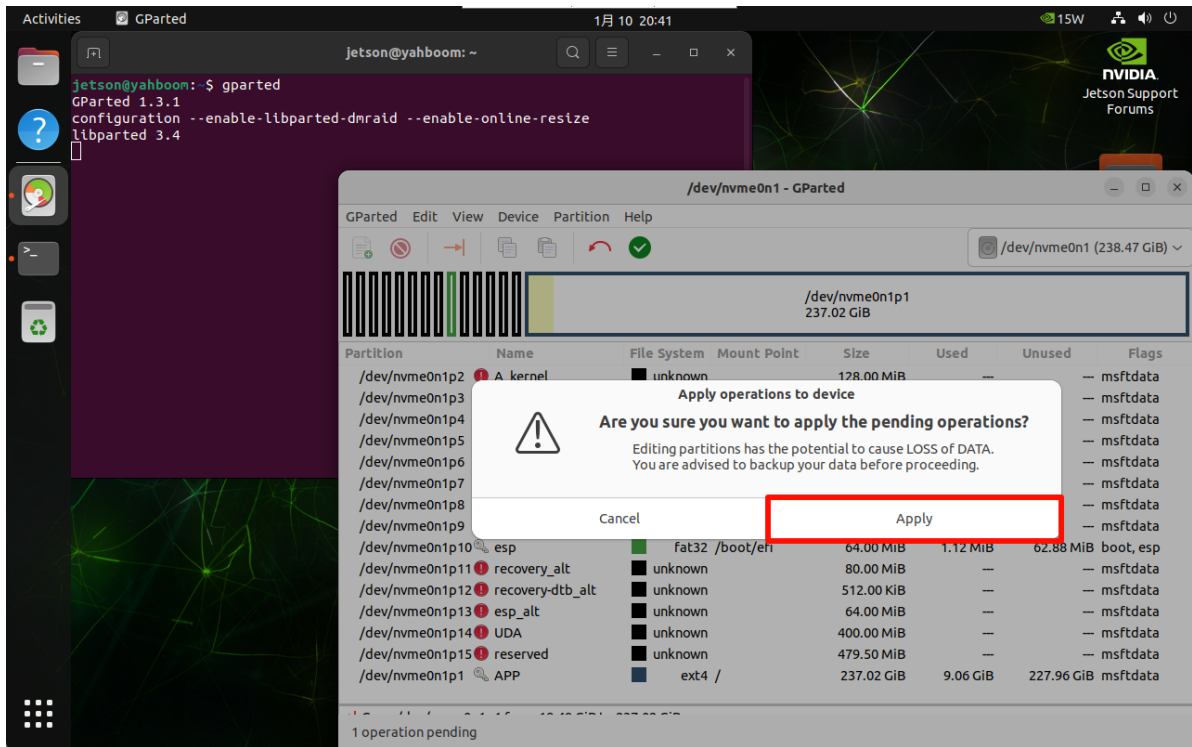


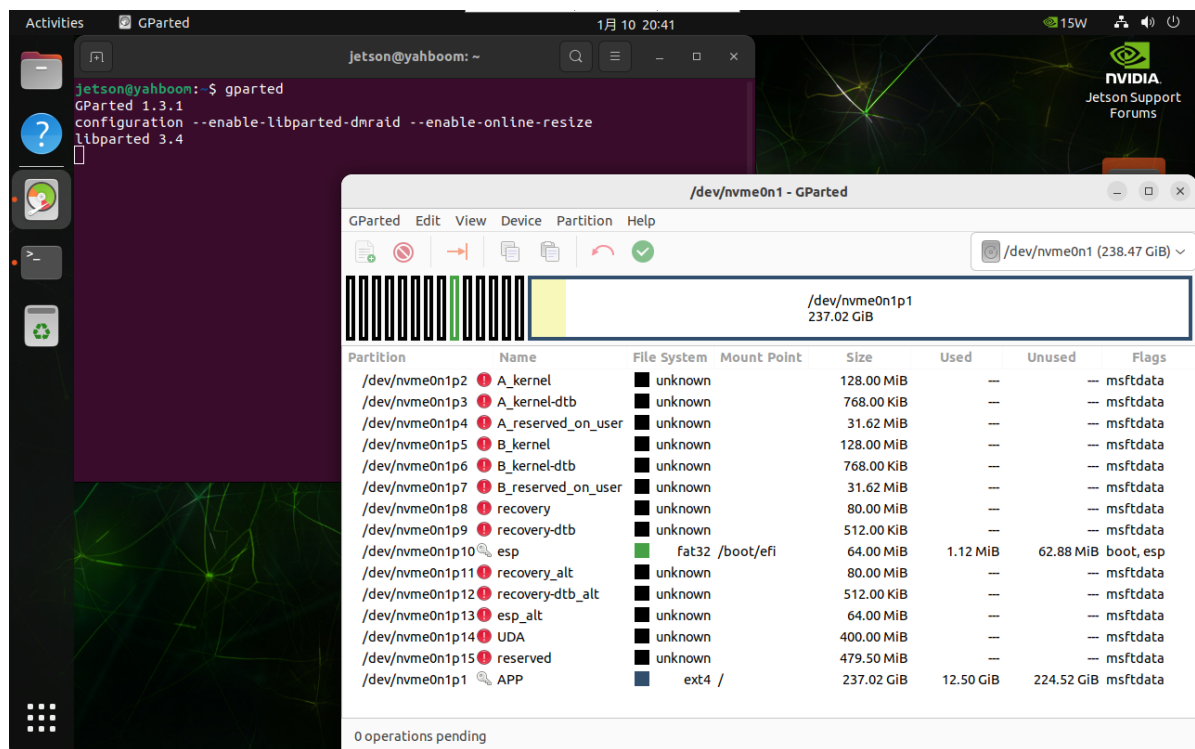
You can adjust the partition size by sliding the slider to the right to maximize the space.



Confirm the partition adjustment operation.







After the partitioning is completed, close the `GParted` software.